

Technology - Taiwan

Thoughts on Rubin Ultra spec delay; higher confidence on testing interface/substrate

Industry Overview

Share price likely hit by delay/cancellation of certain specs

Share prices were weak (down 8-9% on average) on July 6th across the subsectors of CCL/PCB/substrate and testing interface within our coverage, due likely to the concerns surrounding the delay (or even cancellation) of the Kyber rack and CPO adoption from Rubin Ultra to Feynman platform, as well as the adoption of 2-compute die (rather than 4-compute die) structure of GPU. Although such concerns are valid from the technology perspective considering the constraint from CCL material, warpage issue of big-sized substrates, and the reticle size in AP, we see the share price correction as an enhanced entry point for Buy-rated EMC, TUC, Unimicron, Kinsus, NYPCB, MPI and WinWay in the group, as we consider the AI server capex to remain in the upcycle toward 2028.

Demand scale-down amidst ongoing supply constraint

On CCL/PCB/substrate, we see the potential delay/cancellation as a demand scale-down in response to the supply constraint, which is analogous to the concern we saw in (see report) [DRAM for Vera CPU](#). Yet, we flag a likely steadier demand growth for CCL/PCB/substrate vs other parts (e.g. memory) used in AI server system given their lower price elasticity, grounded by a smaller mix (LSD/MSD %) out of total BoM costs. Further, we believe (see reports) [high-end CCL](#) (M8-grade & above) and [ABF substrates](#) will remain in under-supply until at least end-2027, given the lengthy equipment lead-time vs more structural demand growth driven by server CPU/accelerators.

Still expect Q-based M9 to penetrate faster vs PTFE

Although both quartz-based and PTFE-based CCL are considered to enhance the Dk and Df, the production difficulty of each solution is located on PCB and CCL side respectively. Such difference makes us expect a higher chance for Q-glass based M9-grade CCL to penetrate faster in AI server system, along with the improving yield rate on PCB side. In near/mid-term, we also flag demand upside (or wider S/D gap) for high-end CCL, if the improvement of PCB yield rate (based on M9-Q glass) remains mild.

Pecking order: testing interface/substrate, CCL then PCB

In our view, the subsectors being more immune to the inflection of AI server upcycle will be testing interface/substrate, CCL, PCB by pecking order, as the former ones are closer to the chip-level solution with spec advancement. For CCL, we flag TUC's bigger upside in m/s and pricing in N-T/M-T, vs EMC's chance from substrate material and scarcity based on technology advantage (M9-Q) in L-T. For substrate and testing interface, the content downside (from 2-die solution) could be mitigated by unit upside. We also see the worry over substrate price cuts (by memory customers) is unwarranted given the shrinking BT substrate supply, elevated costs of glass cloth vs strong memory demand.

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CCL – copper clad laminate
PCB – printed circuit board
AP – advanced packaging
CPO – co-packaged optics
GPU – graphic processing unit
EMC – Elite Material
TUC – Taiwan Union
NYPCB – Nan Ya Printed Circuit Board
AI – artificial intelligence
PTFE – polytetrafluoroethylene
DRAM – dynamic random access memory
CPU – central processing unit
BoM – bill of material
ABF – Ajinomoto build-up film
Dk – dielectric constant
Df – dissipation factor
S/D – supply/demand
LSD/MSD – low/mid single-digit
N-T/M-T – near/mid-term
BT – bismaleimide triazine
L-T – long-term

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Refer to important disclosures on page 7 to 10. Analyst Certification on page 4. Price

Exhibit 1: Table of recommendation

Stocks mentioned with tickers in this report

BofA ticker	Ticker	Company name	Price (LC)	Rating
ETMCF	2383 TT	EMC	NT\$ 5475	C-1-7
TWUNF	6274 TT	TUC	NT\$ 1630	C-1-7
UMCRF	3037 TT	Unimicron	NT\$ 917	C-1-7
KNSUF	3189 TT	Kinsus	NT\$ 763	C-1-7
NANYF	8046 TT	NYPGB	NT\$ 1075	C-1-7
XMJCF	6223 TT	MPI	NT\$ 6855	C-1-7
XWCLF	6515 TT	WinWay	NT\$ 8915	C-1-7

Source: BofA Global Research, company data

BofA GLOBAL RESEARCH

Price objective basis & risk**Elite Material (ETMCF)**

Our PO of NT\$7,800 is based on 34x 2H27-1H28E P/E, at the high-end of its historical range of 6-34x since 2016. In our view, the multiple is appropriate, given the market-share expansion in server/networking, where we see structural growth in addressable market.

Upside risks are (1) share gain at substrate CCL, (2) stronger 5G smartphone shipment, (3) greater 5G smartphone spec upgrades, and (4) favorable FX and raw material cost change.

Downside risks are (1) share loss at iPhone, (2) keener competition in high-end CCL, and (3) unfavorable FX and raw material cost change.

Kinsus (KNSUF)

We set our PO at NT\$1,070 based on 35x 2H27-1H28E P/E, which is at mid-to-high-end level in its historical trading range (8-40x) since 2013. In our view, the multiple is backed by ROE and operating margin recovery to 30% and 23% in 2027-28E.

Upside risks to our PO are (1) tighter ABF supply, (2) increasing adoption of SiP and AiP, (3) lower competition pressure and (4) slow development of alternative technology.

Downside risks are (1) greater ABF expansion from peers, (2) muted SiP/AiP adoption, (3) more severe competition, and (4) unfavorable technology development.

MPI Corporation (XMJCF)

Our PO NT\$8,250 is based on 40x 2028E P/E, which is at the high-end of the company's historical trading range (10x-40x). In our view, the valuation multiple is underpinned by 1) 77% earnings CAGR in 2026-28E, 2) operating margin of 43-44% in 2027-28E, 3) ROE of 55-56% in 2027-28E, 4) net cash of NT\$33/share, and 5) free cash flow generation of NT\$180/share in 2027-28E.

Downside risks to our PO are:

- 1) slower end market demand, which may undermine the company's revenue growth and operating leverage
- 2) customers' switch from VPC probe card to MEMS solution, where MPI has a less dominant position
- 3) keener competition especially from its foreign competitors, which may dampen the margin profile
- 4) failure to execute or delay of in-sourcing strategy on PCB, which may limit the firm's delivery/supply to customers.

NYP CB (NANYF)

Our PO of NT\$1,430 is based on 35x 2H27-1H28E P/E, which in our view, the multiple backed by ROE and operating margin expansion to 37% and 30% in 2027-28E, as well as a net cash (to equity) position of 21%.

Upside risks to our PO are (1) tighter ABF supply, (2) higher SiP/AiP adoption, and (3) less intensified competition pressure in PCB business.

Downside risks are (1) greater ABF expansion from peers, (2) muted SiP/AiP adoption, and (3) more severe competition in PCB business.

Taiwan Union Technology Corporation (TWUNF)

Our PO NT\$2,600 is based on 30.5x 2H27-1H28E P/E, which is at the high-end of the historical trading range of 8-31x during 2016-2026. In our view, the valuation multiple is justified, given 1) gross/operating margin expansion to 29% and 22-23% in 2026-28E, 2) expanding ROE toward around 53% in 2026-28E, and 3) robust free-cash flow generation in 2027-28E.

Downside risks are

- 1) Fluctuation in raw material prices, which could lead to a more volatile gross margin for TUC in the near term
- 2) More intensified competition in CCL, which could pressure TUC's overall margin profile
- 3) Adoption of new type of upstream material (e.g., quartz), which could bring business uncertainty for the names in CCL industry, including TUC
- 4) Worse-than-expected execution by the company, which could cause market-share loss and/or decreased business traction with customers

Unimicron (UMCRF)

Our PO of NT\$1,400 is based on 33x 2H27-1H28E, which is roughly 1x standard deviation above the historical average of 20x since 2017. In our view, the multiple is appropriate as we expect Unimicron earnings to be supported by its high-end technology offerings, where end market demand seems more resilient.

Upside risks are (1) tighter ABF supply, (2) greater PC and regular server demand, and (3) lower competition pressure.

Downside risks are (1) greater ABF expansion from peers, (2) lower ASP content increase at AI server, and (3) bigger drop in the demand of consumer electronic products.

WinWay Technology (XWCLF)

Our PO NT\$11,800 is based on 37x 2H27-1H28E P/E, which is at mid-to-high-end of the company's historical trading range (9-54x). In our view, such valuation multiple can be supported by 1) operating margin expansion to 40%, 2) ROE expansion to around 71%, 3) free cash flow generation of NT\$316/share in 2027-28, 4) net cash/share of around NT\$31, and 5) its meaningful revenue exposure (higher than 50%) to AI/HPC applications, where the growth is more structural.

Downside risks are:

- 1) slower capex growth by datacenter/hyperscale cloud service providers (CSPs), which will undermine the demand of AI GPU/ASIC and correspondingly pose a negative impact on the consumption of WinWay's testing socket
- 2) less satisfactory execution in the capacity expansion (especially when the pins used in testing socket remain under-supply), which could be resulted from energy shortage amidst the geopolitical tensions and could interrupt the delivery by WinWay and undermine its track record to the customers
- 3) a shortfall in customers' demand or share loss in key customers, which could bring

significant impact to WinWay given its concentrated revenue exposure to top 3 customers

4) a business model change in probe card business which makes WinWay being bypassed, especially when the company still lacks the key technology know-how in pins (used in probe card).

Analyst Certification

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Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
BUY				
	Alchip	ALCPF	3661 TT	Haas Liu
	AMEC	XRTOF	688012 CH	Dai Shen
	ASE Technology Holding	XSRIF	3711 TT	Haas Liu
	ASE Technology Holding -ADR	ASX	ASX US	Haas Liu
	ASMPT	ASMVF	522 HK	Simon Woo, CFA
	Aspeed	XLKMF	5274 TT	Mike Yang
	Chroma ATE	CRMJF	2360 TT	Haas Liu
	eMemory	XYLWF	3529 TT	Mike Yang
	GigaDevice	XXGIF	603986 CH	Daley Li, CFA
	Global Unichip Corp.	GBUHF	3443 TT	Haas Liu
	Grand Process Technology Corp	XZOWF	3131 TT	Haas Liu
	Hanmi Semiconductor	HNSIF	042700 KS	Simon Woo, CFA
	Hon Precision	XXPIF	7769 TT	Haas Liu
	Horizon Robotics	HRZRF	9660 HK	Daley Li, CFA
	Ingenic	XISCF	300223 CH	Dai Shen
	InnoScience Technology	XSCHF	2577 HK	Daley Li, CFA
	JCET Group Co Ltd	XJIEF	600584 CH	Dai Shen
	King Yuan Electronics Corp.	KYUFF	2449 TT	Haas Liu
	Kinik	KIKCF	1560 TT	Haas Liu
	MediaTek	MDTKF	2454 TT	Haas Liu
	Montage Technology	XRDF	688008 CH	Daley Li, CFA
	MPI Corporation	XMJCF	6223 TT	Mike Yang
	Nanya Technology	NNYAF	2408 TT	Simon Woo, CFA
	NCE Power	XVFFF	605111 CH	Daley Li, CFA
	OmniVision	XXHQF	603501 CH	Dai Shen
	Phison Electronics	PISNF	8299 TT	Simon Woo, CFA
	Powerchip Semiconductor Manufacturing Co	XCHPF	6770 TT	Haas Liu
	Powertech Technology	XPPZF	6239 TT	Simon Woo, CFA
	Rockchip	XRPXF	603893 CH	Daley Li, CFA
	Samsung C&T	XCDTF	028260 KS	Simon Woo, CFA
	Samsung Elec -G	SSNHZ	SMSN LI	Simon Woo, CFA
	Samsung Electronics	SSNLF	005930 KS	Simon Woo, CFA
	Samsung Electronics Preferred	SSNPF	005935 KS	Simon Woo, CFA
	Silergy Corp.	SLEGF	6415 TT	Mike Yang
	Silicon Motion	SIMO	SIMO US	Simon Woo, CFA
	SK Square	SKSQF	402340 KS	Simon Woo, CFA
	Taiwan Semiconductor Manufacturing Co.	TSM	TSM US	Haas Liu
	Taiwan Semiconductor Manufacturing Co.	TSMWF	2330 TT	Haas Liu
	Vanguard International Semiconductor Co	VGILF	5347 TT	Haas Liu
	Winbond Electronics	WBEKF	2344 TT	Dai Shen
	WinWay Technology	XWCLF	6515 TT	Mike Yang
	WT Microelectronics	XZOPF	3036 TT	Mike Yang
NEUTRAL				
	ASMedia Technology Inc.	XZSFF	5269 TT	Mike Yang
	Black Sesame Intl Holding	BSIHF	2533 HK	Daley Li, CFA
	Crystal Clear	XPPTF	300655 CH	Dai Shen
	Hangzhou Silan Microelectronics	XDFRF	600460 CH	Daley Li, CFA
	LX Semicon	XLXSF	108320 KS	Simon Woo, CFA
	Novatek	NVKMF	3034 TT	Haas Liu
	Realtek	RLTKF	2379 TT	Mike Yang
	Shenzhen Goodix	XQPLF	603160 CH	Daley Li, CFA
UNDERPERFORM				
	Faraday	FDYTF	3035 TT	Haas Liu
	GlobalWafers	XWLFF	6488 TT	Mike Yang
	Hua Hong Semi	HHUSF	1347 HK	Dai Shen
	Lion Electronics	XDHFF	605358 CH	Dai Shen
	M31 Technology	XMTZF	6643 TT	Mike Yang
	Macronix International	MXICF	2337 TT	Dai Shen
	Maxscend	XMVSF	300782 CH	Daley Li, CFA
	Parade	PRDWF	4966 TT	Mike Yang
	Soulbrain	XSBOF	357780 KS	Simon Woo, CFA
	United Microelectronics Corp.	XUMIF	2303 TT	Haas Liu
	United Microelectronics Corp.	UMC	UMC US	Haas Liu

APR - Semiconductor Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	VeriSilicon	XMLZF	688521 CH	Daley Li, CFA
	Win Semiconductors	WSMIF	3105 TT	Haas Liu
	Wonik IPS	XRHQF	240810 KS	Simon Woo, CFA

RSTR

	SK Hynix	HXSCF	000660 KS	Simon Woo, CFA
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APR - Technology Hardware Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
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BUY

	AAC Technologies	AACAF	2018 HK	Katherine Zhu
	Asia Vital Components	AVMPF	3017 TT	Doris Kao
	BizLink	BIZLF	3665 TT	Doris Kao
	BOE Technology Group Co., Ltd	XCJQF	000725 CH	Brad Lin
	Chenbro	CEOMF	8210 TT	Doris Kao
	Cowell	CWLLF	1415 HK	Katherine Zhu
	Crystal-Optech	XMKAF	002273 CH	Katherine Zhu
	Delta Electronics Inc.	DLTEF	2308 TT	Robert Cheng
	Duksan Neolux	XDSXF	213420 KS	Simon Woo, CFA
	Dynamic Holding	XLHCF	3715 TT	Mike Yang
	E Ink Holdings	PWVIF	8069 TT	Doris Kao
	Elite Material	ETMCF	2383 TT	Mike Yang
	Foxconn Industrial Internet	XDAZF	601138 CH	Robert Cheng
	Goertek	XGKCF	002241 CH	Katherine Zhu
	Gold Circuit	GOCCF	2368 TT	Doris Kao
	Hana Microelectronics	XHNNF	HANA TB	Suppamong Iemkongaek, CFA, CMT
	Hon Hai Precision Industry	HNHAF	2317 TT	Robert Cheng
	Huaqin	XHUAF	603296 CH	Katherine Zhu
	Huaqin	XPEZF	3296 HK	Katherine Zhu
	ISU Petasys	IPTYF	007660 KS	Matt Shin
	Jentech	XVFSF	3653 TT	Doris Kao
	King Slide	KGSLF	2059 TT	Doris Kao
	Kinsus	KNSUF	3189 TT	Mike Yang
	Lens Tech	XLTVF	6613 HK	Katherine Zhu
	LG CNS	XLGLF	064400 KS	Simon Woo, CFA
	LG Electronics	LGEAF	066570 KS	Simon Woo, CFA
	LG Innotek	XLGQF	011070 KS	Simon Woo, CFA
	Lotes	ZYZTF	3533 TT	Doris Kao
	Luxshare	XNJQF	002475 CH	Robert Cheng
	NYPFB	NANYF	8046 TT	Mike Yang
	Quanta Computer	QUCPF	2382 TT	Robert Cheng
	Samsung Electro-Mechanics	SMSGF	009150 KS	Simon Woo, CFA
	Samsung SDI	SSDIF	006400 KS	Simon Woo, CFA
	Samsung SDS	XWPBF	018260 KS	Simon Woo, CFA
	Shennan Circuits	XAMKF	002916 CH	Katherine Zhu
	Taiwan Union Technology Corporation	TWUNF	6274 TT	Mike Yang
	Unimicron	UMCRF	3037 TT	Mike Yang
	Universal Display	OLED	OLED US	Simon Woo, CFA
	Victory Giant	XENEF	300476 CH	Katherine Zhu
	Wistron	WICOF	3231 TT	Doris Kao
	Wiwynn	XWQF	6669 TT	Robert Cheng
	Xiaomi Corporation	XIACF	1810 HK	Robert Cheng
	Yageo	YGEQF	2327 TT	Robert Cheng
	Zhen Ding Tech	XFAGF	4958 TT	Doris Kao
	Zhongji Innolight	XZICF	300308 CH	Katherine Zhu
	ZTE Corporation	SHZZF	000063 CH	Katherine Zhu
	ZTE Corporation	ZTCOF	763 HK	Katherine Zhu

NEUTRAL

	Advantech	ADTEF	2395 TT	Robert Cheng
	Asustek	AKCPF	2357 TT	Robert Cheng
	AU Optronics	AUOTF	2409 TT	Brad Lin
	AU Optronics	AUOTY	AUOTY US	Brad Lin

APR - Technology Hardware Coverage Cluster

Investment rating	Company	BofA Ticker	Bloomberg symbol	Analyst
	Auras Technology	AUTCF	3324 TT	Doris Kao
	BYD Electronic	BYDIF	285 HK	Katherine Zhu
	DS Precision	XBASF	002384 CH	Katherine Zhu
	Innolux Corporation	CMLXY	3481 TT	Brad Lin
	Largan Precision	LGANF	3008 TT	Robert Cheng
	Lenovo Group	LNVGf	992 HK	Robert Cheng
	Lenovo Group	LNVGy	LNVGy US	Robert Cheng
	Lite-On Tech	LOTZF	2301 TT	Doris Kao
	Sunny Optical	SNPTF	2382 HK	Katherine Zhu
	TCL Technology Group Corp	XTOOF	000100 CH	Brad Lin

UNDERPERFORM

	Acer	ASIYF	2353 TT	Robert Cheng
	Catcher Tech	CHERF	2474 TT	Robert Cheng
	Compal Electron	XLCPF	2324 TT	Doris Kao
	Delta Electronics (Thailand)	XLETF	DELTA TB	Suppamong lemkongaek, CFA, CMT
	Everwin Precision	XSHZF	300115 CH	Katherine Zhu
	Genius Electronic Optical	GNSEF	3406 TT	Robert Cheng
	Lens Tech	XLTGF	300433 CH	Katherine Zhu
	LG Display	LPHLF	034220 KS	Simon Woo, CFA
	LG Display-ADR	LPL	LPL US	Simon Woo, CFA
	Lianchuang Electronic	XWOFF	002036 CH	Katherine Zhu
	Pegatron	PGTRF	4938 TT	Robert Cheng
	Sanan Opto	XEQYF	600703 CH	Katherine Zhu
	Seoul Semiconductor	SLSOF	046890 KS	Simon Woo, CFA
	Shengyi	DGUNF	600183 CH	Mike Yang
	Sunway Communications	XWSVF	300136 CH	Katherine Zhu
	Tianma	XVZLF	000050 CH	Brad Lin
	Transsion	XCINF	688036 CH	Robert Cheng

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Buy	34	55.74%	Buy	16	47.06%
Hold	10	16.39%	Hold	3	30.00%
Sell	17	27.87%	Sell	6	35.29%

Equity Investment Rating Distribution: Technology Group (as of 30 Jun 2026)

Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
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Hold	87	21.70%	Hold	45	51.72%
Sell	71	17.71%	Sell	21	29.58%

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Coverage Universe	Count	Percent	Inv. Banking Relationships ^{R1}	Count	Percent
Buy	1987	56.11%	Buy	1190	59.89%
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Sell	757	21.38%	Sell	391	51.65%

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